MADRAS INSTITUTE OF TECHNOLOGY CAMPUS, ANNA UNIVERSITY

DEPARTMENT OF COMPUTER TECHNOLOGY

CS6811 - PROJECT WORK

THIRD REVIEW

AutoML-Driven Predictive Analysis for Insights from Analytic Survey Data

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Domain: Artificial Intelligence, Machine Learning, Social Network Analysis.

I. INTRODUCTION

Machine Learning (ML) is a technology that allows computers to learn from data and improve their performance over time without being explicitly programmed. It powers many applications we use every day, such as virtual assistants, recommendation systems, and fraud detection tools. At its core, ML involves training algorithms on data to find patterns and make predictions. However, setting up a successful ML workflow involves multiple steps like collecting data, cleaning it, choosing the right model, and fine-tuning it—all of which can be complex and require significant time and expertise.

To make this process easier and more accessible, Automated Machine Learning (AutoML) has emerged as a solution. AutoML tools aim to automate the end-to-end process of applying machine learning to real-world problems, allowing even non-experts to build high-performing models. However, AutoML still faces several limitations. Many tools are mainly designed for specific types of data (like tabular data), and may struggle with more complex or multi-modal datasets involving text, images, or audio. Additionally, AutoML systems can act like ”black boxes,” offering limited transparency into how decisions are made, and they may not always adapt well to changing data or real-time feedback. Despite these challenges, AutoML continues to grow and evolve, bringing us closer to more intelligent and accessible AI systems.

II. PROBLEM STATEMENT

The increasing demand for machine learning (ML) solutions across industries is often hindered by the time-consuming and error-prone manual tasks involved in the early stages of ML development. Data scientists spend considerable effort on dataset loading, exploratory data analysis (EDA), and data preparation, which can slow progress and introduce inconsistencies. This project aims to address these challenges by introducing a semi-automated system that streamlines the initial ML workflow. It supports easy ingestion of datasets in common formats like CSV and Excel and standardizes them into a unified structure for downstream compatibility. The system also automates the creation of interactive EDA reports to quickly reveal data quality, distributions, correlations, and other key insights. This reduces manual effort, improves data consistency, and enhances understanding early in the process. Ultimately, the system establishes a solid foundation for expanding into a full AutoML pipeline, including model selection, training, and adaptive learning.

III. LITERATURE SURVEY

S. No.	Paper	Journal/Transactions	System Proposed	Limitations
1	"Transitioning from MLOps to LLMOps: Navigating the Unique Challenges of Large Language Models" Authored By, S. Pahune, Z. Akhtar	Department of Networks and Computer Security, SUNY Polytechnic Inst., Utica, NY, USA, p.36 ResearchGate **(2025)**	• This paper compares LLMOps (Large Language Model Operations) with MLOps, focusing on the unique challenges, infrastructure needs, and methodologies for managing LLM lifecycles, such as fine-tuning, deployment, and monitoring.	• Complex scalability requirements for LLMs in production environments.
2	"Meta-Learning in Neural Networks: A Survey" Authored By, T. Hospedales, A. Antoniou, P. Micaelli, A. Storkey	IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 46, no. 7, pp. 4763-4779 **(2024)**	• A comprehensive survey on meta-learning, introducing a new taxonomy to categorize current methods and exploring its applications in few-shot learning and reinforcement learning.	• Accurately categorizing diverse meta-learning methods. • Struggles with complex multi-modal task distributions.
3	"Constructing a Meta-Learner for Unsupervised Anomaly Detection" Authored By, M. Gutowska, S. Little, A. Mccarren	IEEE Access, vol. 11, pp. 45815-45825 **(2023)**	• A meta-learning approach selecting the best anomaly detection algorithm based on meta-features from unlabelled data, outperforming existing methods.	• Limited generalization across all anomaly detection tasks. • Reliance on high-quality meta-features for accurate selection.
4	"Model Selection for Generic Contextual Bandits" Authored By, A. Ghosh, A. Sankararaman, K. Ramchandran	IEEE Transactions on Information Theory, vol. 70, no. 1, pp. 656-675 **(2024)**	• An adaptive algorithm selects models for contextual bandits, minimizing selection costs while offering sharper guarantees in linear contexts.	• Higher model selection cost in ETC compared to ACB.
5	"Performance Indicator-Based Adaptive Model Selection for Offline Data-Driven Multi-objective Evolutionary Optimization" Authored By, Z. Liu, H. Wang, Y. Jin	IEEE Transactions on Cybernetics, vol. 53, no. 10, pp. 6263-6276 **(2023)**	• An adaptive surrogate-assisted evolutionary algorithm used for switching between Kriging models and RBFNs based on approximation error.	• Depends only on surrogate model quality.

S. No.	Paper	Journal/Transactions	System Proposed	Limitations
6	"Curiosity-Driven Class-Incremental Learning via Adaptive Sample Selection" Authored By, Q. Hu, Y. Gao, B. Cao	IEEE Transactions on Circuits and Systems for Video Technology, vol. 32, no. 12, pp. 8660-8673 **(2022)**	• A curiosity-driven class-incremental learning approach is proposed to reduce catastrophic forgetting by selectively learning informative samples based on uncertainty and novelty, improving model robustness and balance between new and old knowledge.	• Dependence on accurate quantification of uncertainty and novelty. • Potential inefficiency in highly imbalanced datasets.
7	"A Self-Adaptive Parameter Selection Trajectory Prediction Approach via Hidden Markov Models" Authored By, S. Qiao, D. Shen, X. Wang, N. Han	IEEE Transactions on Intelligent Transportation Systems, vol. 16, no. 1, pp. 284-296 **(2023)**	• A hidden Markov model-based trajectory prediction algorithm (HMTP) is proposed to improve accuracy in network-constrained environments. An enhanced version, HMTP*, introduces self-adaptive parameter selection and a density-based partitioning algorithm for better prediction efficiency.	• Sensitivity to parameter selection and tuning. • Potential inefficiency in highly dynamic environments with abrupt speed variations.
8	"Dynamic Ensemble Selection for Imbalanced Data Streams With Concept Drift" Authored By, B. Jiao, Y. Guo	IEEE Transactions on Neural Networks and Learning Systems, vol. 35, no. 1, pp. 1278-1291 **(2024)**	• A dynamic ensemble selection method (DES-ICD) is proposed for handling concept drift and class imbalance in data streams. It employs AnnSMOTE to balance data and selects base classifiers dynamically based on local performance rather than global accuracy.	• Increased computational complexity due to local classifier selection.
9	"Automatic Title Generation for Text with Pre-trained Transformer Language Model" Authored By, P. Mishra, C. Diwan, S. Srinivasa, G. Srinivasaraghavan	IEEE Transactions on Information Theory, vol. 50, no. 1, pp. 656-675 **(2024)**	• A model that adapts and generalizes using meta-learning for automatic title generation from text.	• Handling complex, multi-modal task distributions. • Mitigating data distribution shifts.
10	"Unsupervised Feature Selection via Controllable Adaptive Graph Learning and Discriminative Feature Learning" Authored By, P. Huang, M. Xie, X. Yang	IEEE Transactions on Neural Networks and Learning Systems, vol. 35, no. 11, pp. 15600-15614 **(2024)**	• Unsupervised feature selection faces challenges in preserving intrinsic structure and finding uncorrelated features. CAG-U and CAG-I use adaptive graph learning and discrete projection to overcome inefficiencies in high-dimensional data.	• Dependence on prior knowledge of subspace dimension. • Inefficiency in handling extremely high-dimensional datasets.

S. No.	Paper	Journal/Transactions	System Proposed	Limitations
11	"Modyn: Data-Centric Machine Learning Pipeline Orchestration" Authored By, M. Böther, T. Robroek, V. Gsteiger, R. Holzinger, X. Ma, P. Tözün	Proc. Int. Conf. Manage. Data (SIGMOD'25), ACM, New York, NY, USA, pp. 1–30 **(2025)**	• Modyn, a data-centric ML platform, enables continuous model training on growing datasets by applying data selection and triggering policies to optimize retraining costs.	• Effectiveness of data selection policies depends heavily on the quality and representativeness of the data.
12	"Advances and Challenges in Meta-Learning: A Technical Review" Authored By, A. Vettoruzzo, M.R. Bouguelia, J. Vanschoren	IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 46, no. 7, pp. 4763-4779 **(2024)**	• A meta-learning system to enhance model adaptation and generalization across diverse tasks, particularly in data-scarce scenarios. It integrates multi-task learning, transfer learning, and continual learning to improve efficiency and performance.	• Adaptation to data distribution shifts in dynamic environments.
13	"Online AutoML: An Adaptive AutoML Framework for Online Learning" Authored By, B. Celik, P. Singh, J. Vanschoren	Machine Learning, vol. 112, no. 6, pp. 1897–1921, doi: 10.1007/s10994-022-06262-0 **(2023)**	• An adaptive Online Automated Machine Learning (OAML) system that continuously optimizes learning pipelines in dynamic environments with evolving data distributions.	• Inefficiencies in handling highly complex or large-scale data streams.
14	"Adaptive Model Selection Framework: An Application to Airline Pricing" Authored By, N. Shukla, A. Kolbeinsson, L. Marla, K. Yellepeddi	Journal of Machine Learning Research (JMLR), arXiv:1905.08874v1 [cs.LG], pp. 1-25 **(2019)**	• An adaptive meta-decision framework using Thompson sampling to dynamically route customer requests to different airline pricing models.	• Assumes a consistent data stream and prior knowledge.
15	"Multitask Learning with Meta-Learning: Optimizing Learning of Multiple Related Tasks" Authored By, A. Ruder, M. E. Peters, M. L. M. Asano	Journal of Machine Learning Research, vol. 20, no. 1, pp. 1897-1921 **(2023)**	• A multitask learning framework optimized through meta-learning, enhancing generalization across a wide array of tasks and datasets.	• High dependence on prior data quality and representativeness for successful multitask learning.

IV. Research Gap Identified

- Existing meta-learning systems struggle to adapt efficiently in highly diverse environments where data distributions change frequently.
- Current meta-learning methods lack precision in handling diverse and complex multi-modal tasks.
- Issues with scalability in large or high-dimensional datasets.
- Performance inefficiencies due to high complexity.

These gaps highlight the need for more adaptive, efficient, and generalizable AI systems that can handle diverse data environments, complex distributions, and scalability challenges.

V. Objectives

- To automate and streamline the early stages of ML workflows by simplifying dataset ingestion, standardization, and exploratory data analysis.
- To enhance data quality and understanding through automated EDA reports that highlight key insights, distributions, and anomalies.
- To establish a scalable foundation for full AutoML integration, enabling future automation of model selection, training, and adaptive learning.

VI. Dataset Description

A. Tabular/Text Dataset

The tabular dataset is obtained from Kaggle and consists of 250 records of Amazon sales transactions(2025), including details about the products sold, customers, payment methods, and order statuses.It includes 11 parameters such as

- **Order ID**: Unique identifier for each order (e.g., ORD0001).
- **Date**:Date of the order.
- **Product**: Name of the product purchased.
- **Category**: Product category (Electronics, Clothing, Home Appliances, etc.).
- **Price**: Price of a single unit of the product.
- **Quantity**:Number of units purchased in the order.
- **Total Sales**: Total revenue from the order (Price × Quantity).
- **Customer Name**: Name of the customer.
- **Customer Location**: City where the customer is based.
- **Payment Method**: Mode of payment (Credit Card, Debit Card, PayPal, etc.).
- **Status**:Order status (Completed, Pending, or Cancelled).

B. Image Dataset

The image dataset is obtained from Kaggle and consists of 32,412 records of information about product listings from an e-commerce platform(2021). It likely originates from a product deduplication or similarity detection task, where the goal is to identify which listings refer to the same actual product based on text and image data.It includes,

- **Posting id**: Unique identifier for each product listing.
- **Image**:Filename of the product image (eg jpg).
- **Image phush**: Perceptual hash (phash) of the image used to compute visual similarity.
- **Title**: Original product title from the e-commerce listing.
- **Label group**: Group ID indicating which listings are duplicates (same product). This is the target label for training.
- **title prep**:Cleaned and preprocessed version of the title, e.g., lowercased, punctuation removed.
- **pos**: Part-of-speech tags for the titleprep (e.g., NN = noun, JJ = adjective).
- **word count**: Total number of words in titleprep.
- **char count**: Total number of characters in titleprep.
- **avg word length**: Average number of characters per word.
- **Stop words count**:Count of stopwords present in titleprep.

VII. Block Architecture Diagram

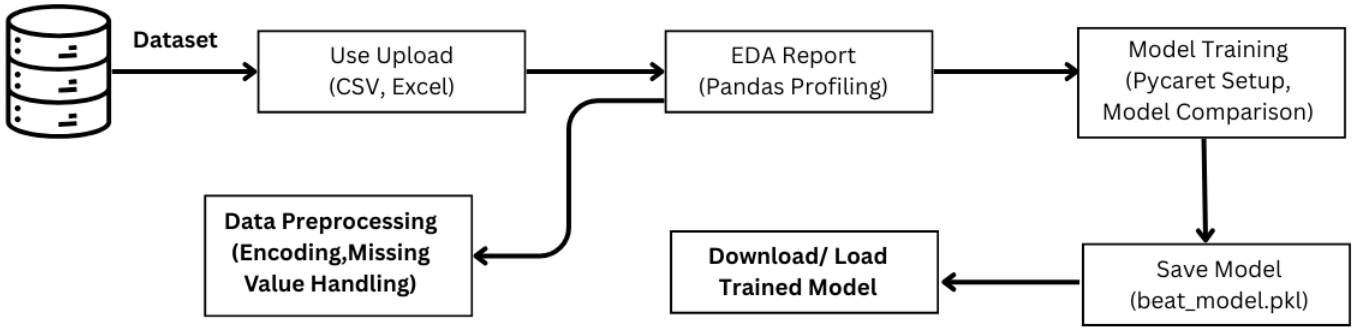


Fig. 1. *Block Architecture diagram of Adaptive Multi-Modal Learning Systems*

VIII. Project Modules and Description

A. Data ingestion and preprocessing

Input: User-uploaded dataset file (.csv or .xlsx) via Streamlit File Uploader.
Processing:

- Validates the uploaded file type and reads it into a Pandas DataFrame.
- Saves the validated DataFrame as source data.csv for later access.
- Standardizes column names and fills missing values in categorical columns with default placeholders.
- Generates an interactive EDA report using the cleaned dataset.

Outcome: Cleaned and saved dataset (source data.csv) and automatically generated EDA Report.

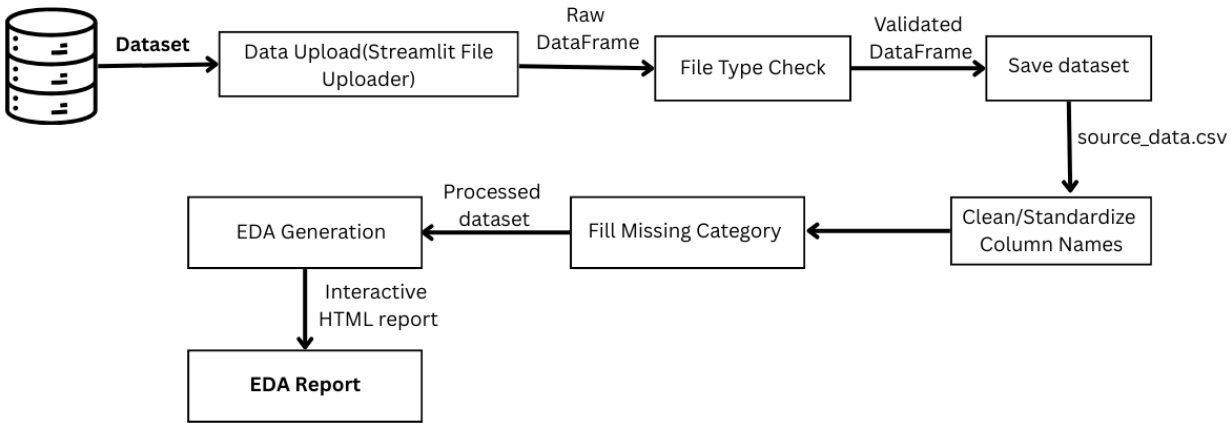


Fig. 2. Architecture Diagram of the Data ingestion and preprocessing Module

B. Model Training and Selection

Input: Processed dataset from EDA stage (numeric, labeled data) and user-selected target variable.
Processing:

- Label encoding is applied to categorical columns to convert them into numeric format.
- Mode imputation is used to handle missing values in scalar fields.
- PyCaret setup is initialized with the processed dataset for automated preprocessing and model comparison.

– Models (e.g., RandomForest, LightGBM) are trained, and the best-performing one is selected based on evaluation metrics.

Outcome: Best model saved for deployment or reuse and performance metric report summarizing model evaluation results (Accuracy, AUC, F1-score, etc.).

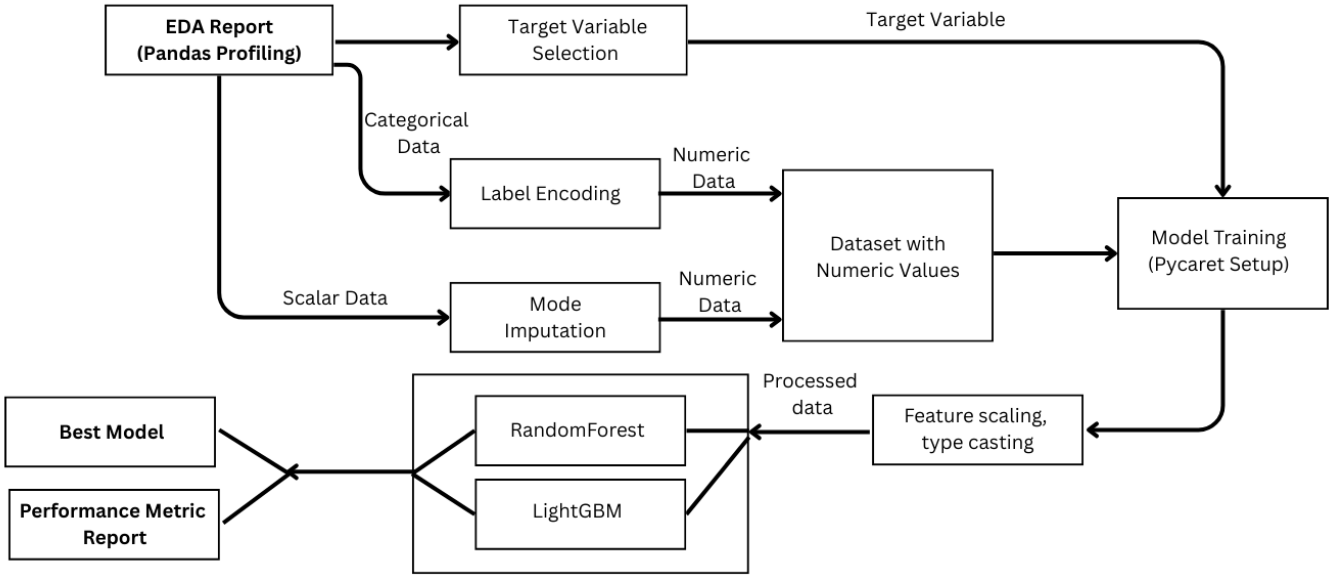


Fig. 3. Architecture Diagram of the Model Training and Selection Module

C. Model comparison

Input: Trained best model (best model.pkl) amd new unseen dataset for inference.

Processing:

– The best-performing model is saved as a .pkl file after training.
– The saved model file is later reloaded for inference tasks.
– A new dataset is provided for generating predictions.
– The loaded model processes the input data and makes predictions based on learned patterns.

Outcome: Predictions for the new dataset.



Fig. 4. Architecture Diagram of Model Comparison Module

Outcome: Best model chosen for deployment.

IX. EXPERIMENTAL SETUP

A. Environment

VS Code, Python 3.9, scikit-learn, XGBoost, LightGBM, CatBoost, NumPy 1.23.5, pandas, ydata-profiling.

B. Dataset

1) Tabular Dataset: A structured dataset of 250 Amazon sales records (2025) from Kaggle, containing product, customer, and transaction details for order status classification.
2) Image Dataset: An e-commerce product listing dataset from Kaggle with 32,412 records (2021), used for duplicate detection based on image and text similarity.

C. Model

- Logistic Regression
- Decision Tree
- Random Forest
- LightGBM
- CatBoost
- k-Nearest Neighbors
- Support Vector Machine (SVM)

D. Evaluation Metrics

1) Accuracy: Accuracy measures the proportion of correctly classified samples over the total number of samples. It is computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

2) Precision, Recall, and F1-Score: These metrics provide detailed insight into model performance, especially for imbalanced datasets.

- **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

- **Recall:**

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

- **F1-Score:**

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

- **Matthews Correlation Coefficient (MCC):** MCC is a robust metric that considers all values from the confusion matrix and is particularly useful for imbalanced classes. It is defined as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{5}$$

- **Training Time (TT):** The time (in seconds or minutes) required to train the model. It is used to compare computational efficiency among models.

3) Confusion Matrix: The confusion matrix summarizes prediction results by comparing predicted and actual labels. It includes true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).
float

X. PSEUDO CODE

Algorithm 1 Dataset Ingestion and EDA Report Generation

```
procedure UPLOAD_AND_EDA(file_uploader, profiler)
    file ← file_uploader.receive_file()
    if file is None then
        return "No file uploaded"
    end if
    if file.type == 'csv' then
        df ← ReadCSV(file)
    else if file.type == 'xlsx' then
        df ← ReadExcel(file)
    else
        return "Unsupported file format"
    end if
    SaveCSV(df, "source_data.csv")
    CleanedDF ← df.copy()
    for each column in CleanedDF.columns do
        if column name is None or empty then
            Replace column name with default name
        end if
    end for
    for each column in CleanedDF do
        if column.type is categorical then
            Replace nulls with "Unknown"
        end if
    end for
    eda_report ← profiler.generate(CleanedDF)
    DisplayProfileReport(eda_report)
    return "EDA report generated successfully"
end procedure
```

Algorithm 2 Model training and selction

```
procedure AUTOMODELTRAINING(df, target)
    df ← HandleMissingValues(df)
    df ← EncodeCategoricals(df)
    df ← RemoveRareClasses(df, threshold = 0.01)
    df ← FeatureScaling(df)
    SetupAutoML(df, target, session_id = 42, feature_selection = True, normalize = True, remove_multicollinearity = True)
    model_list ← ListOfAvailableModels()
    metrics_dict ← EmptyDictionary()
    for all model in model_list do
        trained_model ← TrainModel(model)
        metrics ← EvaluateModel(trained_model, metrics = ["Accuracy", "F1", "AUC"])
        metrics_dict[model] ← metrics
    end for
    best_model ← SelectBestModel(metrics_dict, criterion = "F1")
    final_model ← FinalizeModel(best_model)
    Save(final_model, "best_model.pkl")
end procedure
```

Algorithm 3 LoadModelAndPredict

```
procedure LoadModelAndPredict(new_data)
    best_model ← LoadModelFromDisk("best_model.pkl")
    preprocessing_pipeline ← LoadPreprocessingPipeline("preprocessing_pipeline.pkl")
    new_data_cleaned ← CleanMissingValues(new_data)
    new_data_encoded ← ApplyEncoding(preprocessing_pipeline.encoder, new_data_cleaned)
    new_data_scaled ← ApplyScaling(preprocessing_pipeline.scaler, new_data_encoded)
    expected_features ← GetExpectedFeaturesFromModel(best_model)
    aligned_data ← AlignColumns(new_data_scaled, expected_features)
    predictions ← Predict(best_model, aligned_data)
    readable_predictions ← ConvertPredictionsToLabels(predictions)
    Display(readable_predictions)
end procedure
```

XI. IMPLEMENTATIONS AND OUTPUTS



	Order ID	Date	Product	Category	Price	Quantity	Total Sales	Customer Name	Customer Location	Payment Method	Status
0	ORD0001	14-03-25	Running Shoes	Footwear	60	3	180	Emma Clark	New York	Debit Card	Cancelled
1	ORD0002	20-03-25	Headphones	Electronics	100	4	400	Emily Johnson	San Francisco	Debit Card	Pending
2	ORD0003	15-02-25	Running Shoes	Footwear	60	2	120	John Doe	Denver	Amazon Pay	Cancelled
3	ORD0004	19-02-25	Running Shoes	Footwear	60	3	180	Olivia Wilson	Dallas	Credit Card	Pending
4	ORD0005	10-03-25	Smartwatch	Electronics	150	3	450	Emma Clark	New York	Debit Card	Pending

Fig. 5. Output snippet of the dataset uploaded by the user

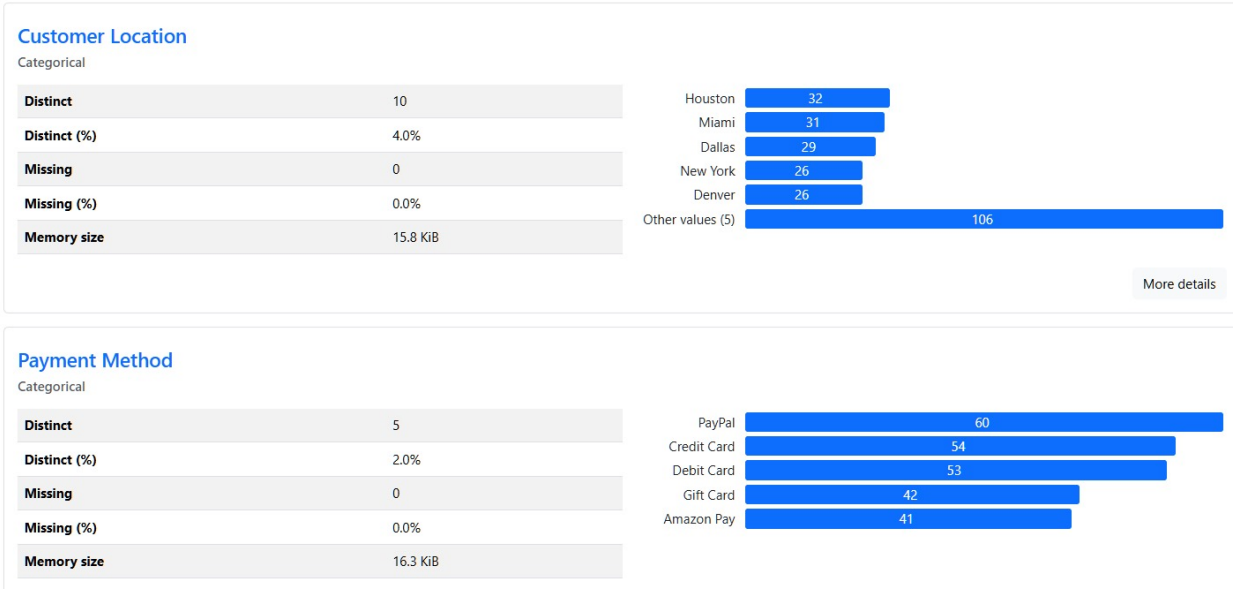


Fig. 6. Output of the EDA report generayed using the pandas-profiling

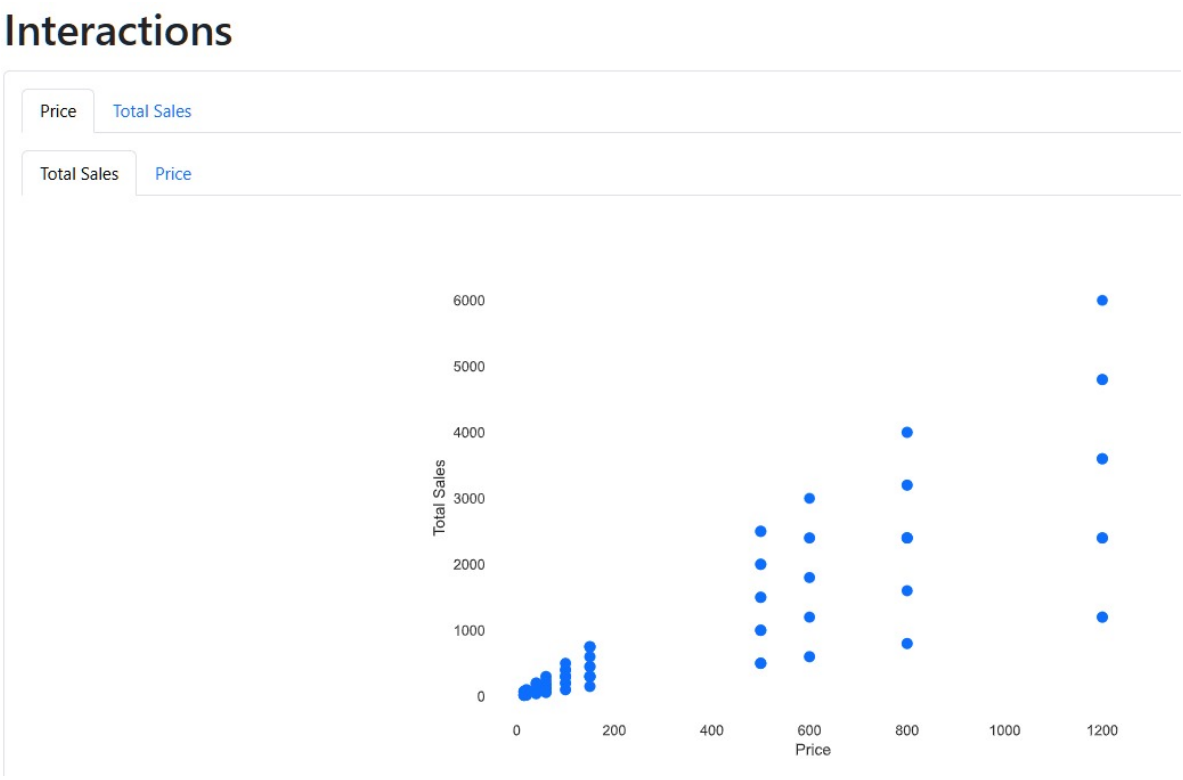


Fig. 7. Output of the model interaction graph between total sales and price

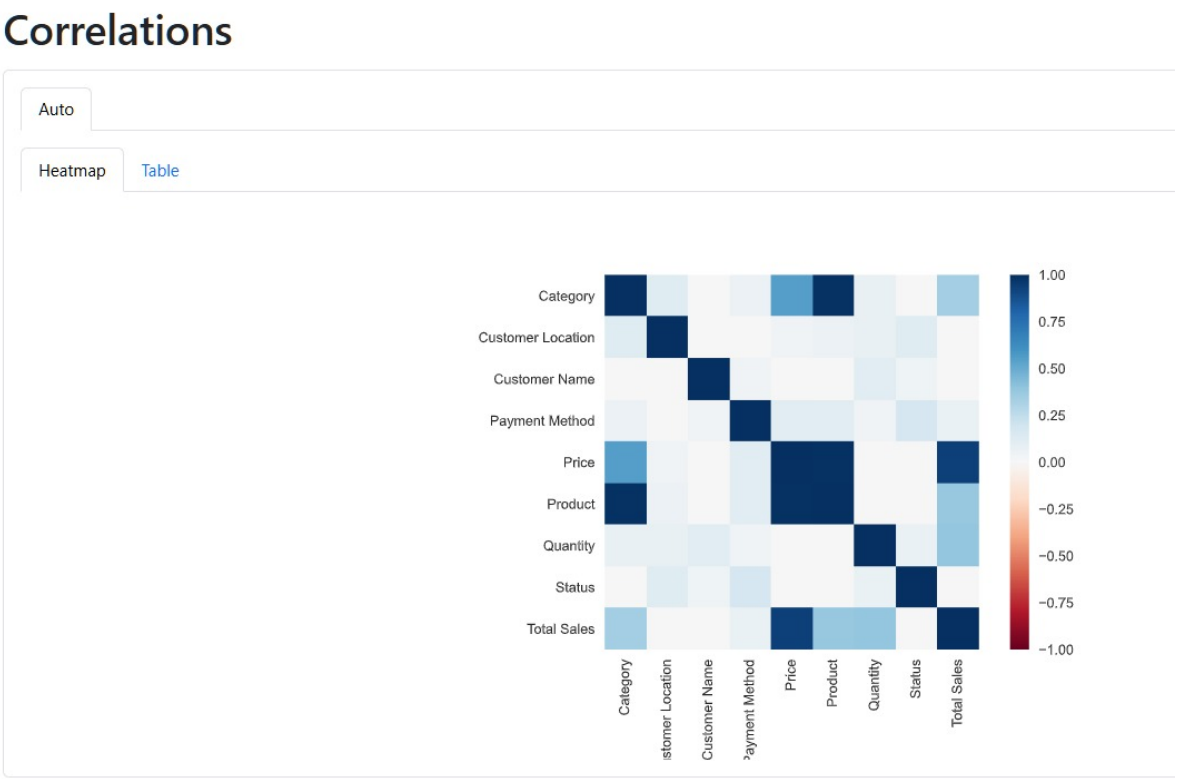


Fig. 8. Output of the confusion matrix for variable interactions

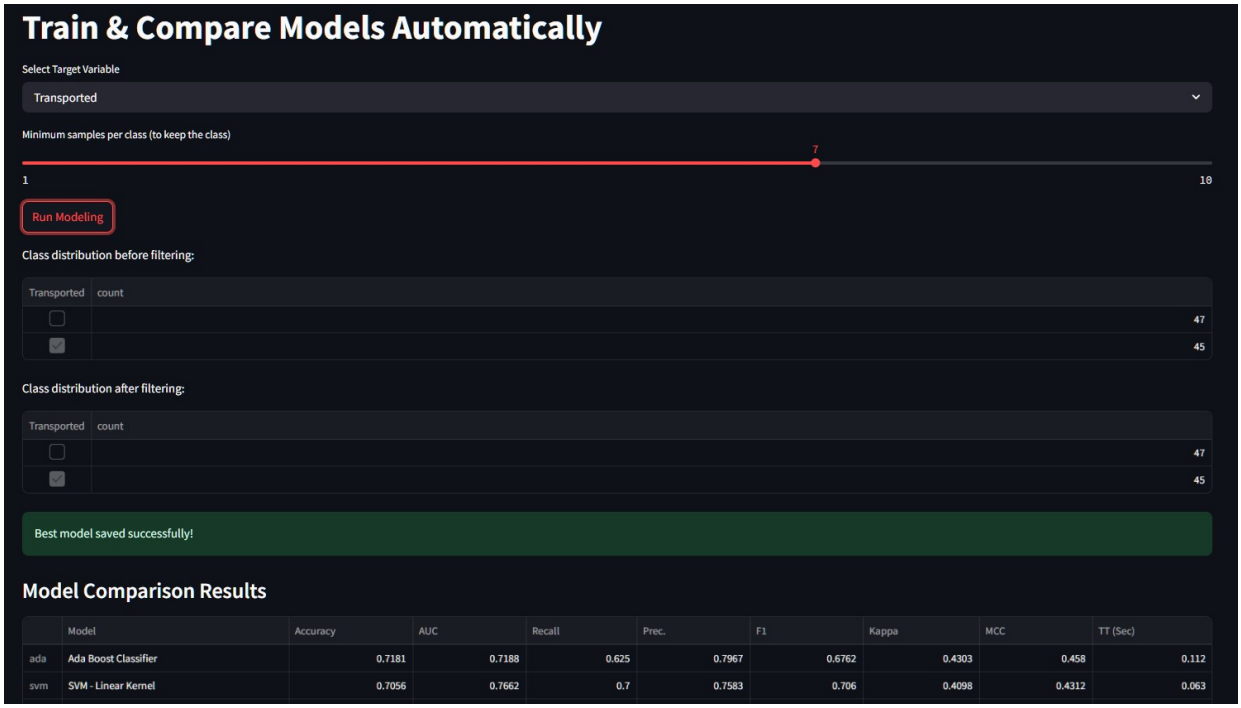


Fig. 9. Output of the model training and best model selection

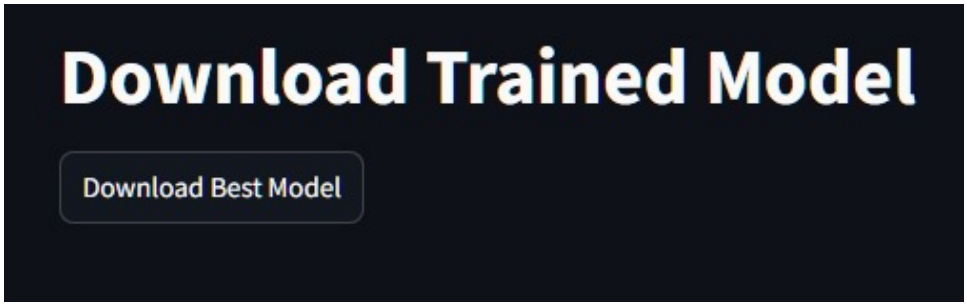


Fig. 10. snippet of UI for downloading the best model for the given dataset

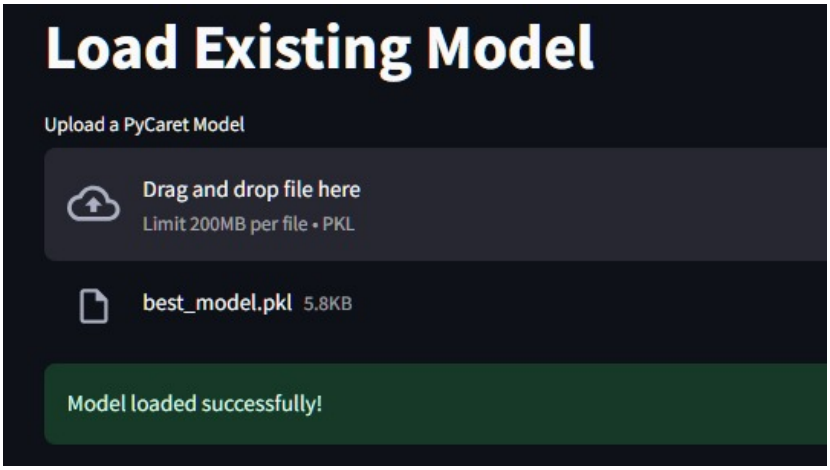
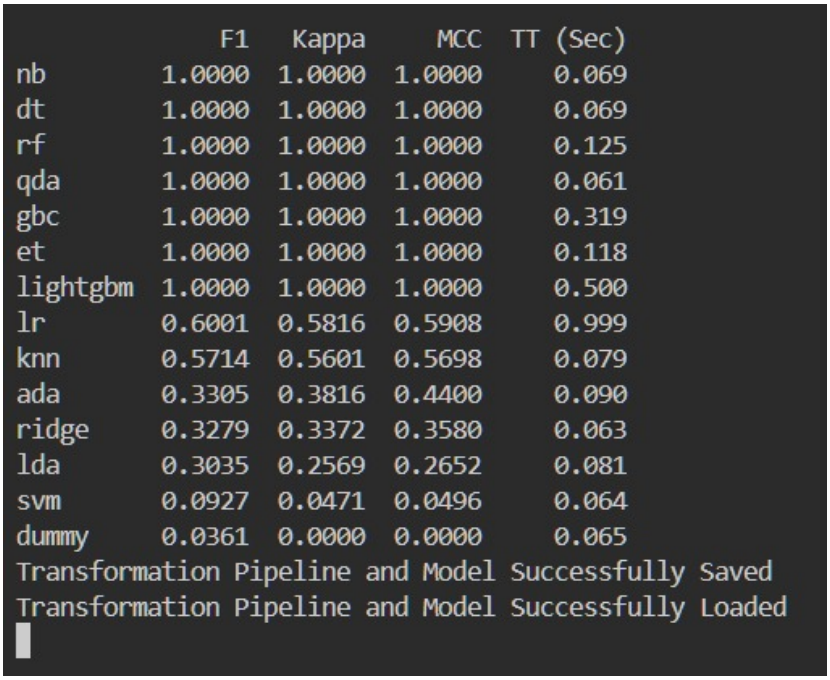


Fig. 11. Snippet of the model loading and evaluation



	F1	Kappa	MCC	TT (Sec)
nb	1.0000	1.0000	1.0000	0.069
dt	1.0000	1.0000	1.0000	0.069
rf	1.0000	1.0000	1.0000	0.125
qda	1.0000	1.0000	1.0000	0.061
gbc	1.0000	1.0000	1.0000	0.319
et	1.0000	1.0000	1.0000	0.118
lightgbm	1.0000	1.0000	1.0000	0.500
lr	0.6001	0.5816	0.5908	0.999
knn	0.5714	0.5601	0.5698	0.079
ada	0.3305	0.3816	0.4400	0.090
ridge	0.3279	0.3372	0.3580	0.063
lda	0.3035	0.2569	0.2652	0.081
svm	0.0927	0.0471	0.0496	0.064
dummy	0.0361	0.0000	0.0000	0.065

Transformation Pipeline and Model Successfully Saved
Transformation Pipeline and Model Successfully Loaded

Fig. 12. Output of the performance metric report for each model

XII. TEST CASES AND MODEL COMPARISON

Test CaseS	Module	Test Description	Expected Outcome
01	Upload	Upload a valid .csv or .xlsx file	File is uploaded and previewed
02	Upload	Upload with missing column names	Columns are auto-renamed and processed
03	Upload	Upload file with missing values	Missing values filled and app continues
04	Profiling	Generate profiling after upload	Shows EDA report using ydata_profiling
05	Profiling	Try profiling without uploading	Warning: "Please upload a dataset first"
06	Model Generation	Train model after selecting target	Best model saved and comparison shown
07	Model Generation	Select target with 1 unique class	Error: "Need at least 2 classes"
08	Model Generation	Handle rare class with one instance	Rare classes removed with info shown
09	Download	Download before training any model	Warning: "No model available"
10	Download	Download after model training	File downloaded as best_model.pkl
11	Load Model	Upload valid PyCaret model (.pkl)	Model loaded with success message
12	Load Model	Upload unsupported file type	Error or warning shown

XIII. PROJECT MODULES AND TASKS

Module	Tasks	Work Status
Data Ingestion	Upload dataset using Streamlit and check file type.	Completed
Data Cleaning	Standardize column names and fill missing categorical values.	Completed
Exploratory Analysis	Generate EDA report using Pandas Profiling.	Completed
Data Preprocessing	Encode labels, impute missing values, scale and cast features.	Completed
Model Training & Eval	Train models (e.g., RandomForest, LightGBM) and evaluate performance.	Completed
Model Deployment	Save the best model and generate predictions on new data.	Completed

TABLE I
OVERVIEW OF MAIN MODULES, TASKS, AND STATUS

REFERENCES

[1] S. Pahune and Z. Akhtar, "Transitioning from MLOps to LLMOps: Navigating the Unique Challenges of Large Language Models," Dept. Netw. Comput. Security, SUNY Polytechnic Inst., Utica, NY, USA, 2025, p. 36 on ResearchGate.

[2] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-Learning in Neural Networks: A Survey," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 9, pp. 5149-5169, Sept. 2022, doi: 10.1109/TPAMI.2021.3079209.

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